

REQUEST FOR INFORMATION



Multi-Entity Autonomy-Enabled Mission Command Reference Architecture

RFI Number: MEAMC-2026-001

Issue Date: March 11, 2026

Response Deadline: April 9, 2026

1. Executive Summary

The Army's Capability Program Executive Mission Autonomy (CPE MA) PM Robotic Control and Integration (PM RCI) is seeking information from industry partners regarding innovative approaches, mature technologies, and implementation strategies for multi-entity autonomous-enabled mission command reference architecture. These systems will enable coordinated operations of heterogeneous autonomous entities (unmanned aerial vehicles, unmanned ground vehicles, unmanned surface/underwater vehicles, and autonomous payloads/sensors) across the full spectrum of military operations. Input and submittals are requested from small businesses, non-traditional firms, and large business concerns, and CPE MA is seeking complete or partial solutions and/or niche concepts or products.

This document serves as a Request for Information (RFI)/Sources Sought for technologies which show potential for meeting the description below. Technologies and solutions may be Commercial Off-the-Shelf (COTS) or developmental items. This is not a Request for Quotation (RFQ), Request for Proposal (RFP) or an Invitation for Bid (IFB), but submittals may form the genesis of future CPE MA requirements. This RFI is a market research tool used to determine the availability and adequacy of potential sources for planning purposes only. Responses to this notice are not considered as offers and cannot be accepted by the Government to form a binding contract. The Government does not intend to award a contract based on responses nor otherwise pay for the preparation of any information presented or the use of any such information. It is plausible that the Government may later issue an RFWP/RFP, depending on mission requirements; thus, submittals must state whether the Company/Entity would likely participate in future proposals for the concepts outlined in their technical write-up.

2. Purpose

The Government seeks to understand:

- Existing and emerging autonomous-enabled mission command reference architectures and frameworks
- Current state-of-the-art multi-entity mission orchestration of autonomous systems
- Technical approaches to addressing key architectural challenges
- Maturity levels and development timelines of solutions for critical capabilities
- Integration approaches with existing systems and platforms
- Verification, validation, and testing methodologies
- Modular Open System Architecture (MOSA) approach and construct

3. Background and Context

Modern military operations will require the coordination of multiple autonomous entities operating across land, air, and maritime domains. Current systems typically employ single-platform autonomy with limited cross-platform coordination capabilities. CPE MA seeks solutions that enable true multi-entity collaboration with graceful degradation in Denied, Degraded, Intermittent, & Limited (DDIL) electro-magnetic spectrum environments.

3.1. Autonomy-Enabled Mission Command Reference Architecture

This aspect of autonomy primarily focuses on planning and collaboration between entities and is an integration layer above and between autonomous effects and autonomous mobility. It involves the systems and algorithms responsible for high-level decision-making, such as route planning, task prioritization, and mission execution. Autonomy-enabled mission command reference architecture enables an autonomous system to operate cohesively with other components, units, and systems, whether human or machine, by interpreting mission objectives, generating plans, and adjusting those plans dynamically based on changes in the environment or mission priorities. Collaboration is a key feature of autonomy-enabled mission command reference architecture, as it requires robust communication protocols and the ability to synchronize actions across a network of systems. This ensures that the autonomous system contributes effectively to a larger operational framework (Figure 1), whether in isolation or as part of a team. Effective autonomy-enabled mission command reference architecture depends on control and its associated user interface, which must incorporate human factors and enable one-to-many control.

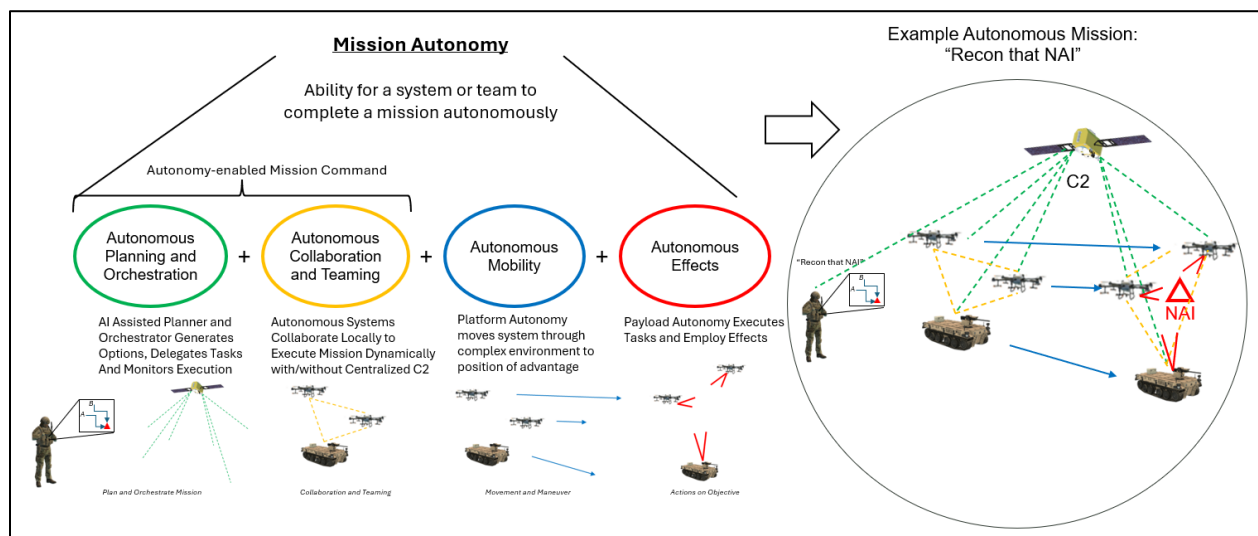


Figure 1: Autonomy-Enabled Mission Command Reference Architecture Framework

3.2. Operational Need

Future conflicts will require:

- Planning and Course-of-Action (CoA) generation for multiple entities
- Coordination of quantities of heterogeneous autonomous entities simultaneously
- Operation in DDIL electro-magnetic spectrum
- Dynamic mission replanning in response to changing battlefield conditions
- Resilience to electronic warfare, cyber-attacks, and kinetic threats
- Meaningful human control/supervision at appropriate decision points
- Other operationally relevant needs as identified by input to this RFI from industry.

4. Reference Architecture Industry Input

CPE MA seeks information for the development of autonomous-enabled mission command reference architecture solutions. The intent of the autonomous-enabled mission command reference architecture is to define a comprehensive framework for developing and implementing autonomous multi-entity systems applicable for military operations. It shall address the coordination of heterogeneous autonomous assets (UAVs, UGVs, USVs, sensors, effectors) operating collaboratively to accomplish complex mission objectives. Please provide information on the use and adaptation of existing reference architectures and/or relevant elements/capabilities that may fit portions of the architecture, and if possible, provide information on the following:

- What open standards (MIL, IETF, ACM, NATO, commercial, other) underpin your autonomous-enabled mission command reference architecture solution?
- What messaging interface(s) do you use between control station and platform? Do you utilize protobuf, efficient XML or other formats to reduce message payload size over constrained/contested networks? Do you use different interfaces for commands / coordination versus mission data exchange?
- Is your autonomous-enabled mission command reference architecture implementation centralized, distributed, hybrid or another structure?
- What are computing requirements for your autonomous-enabled mission command reference architecture implementation as relevant to the user interface and any computation distributed to each unmanned system?
- What modeling and simulation environments do you have for your autonomous-enabled mission command reference architecture implementations?
- What are your wireless transport requirements for your autonomous-enabled mission command reference architecture implementation to be functional and achieve its operational effectiveness?
- What use cases, models or artifacts would it be useful for the USGOV to provide?
- In order to select or define an architecture, CPE MA is considering utilizing a National Advanced Mobility Consortium, Inc. (NAMC) community of practice (COP). Would you recommend an alternative consortium or an alternative approach?
- What is the MOSA approach, construct, and standards that have been implemented and/or recommended for emerging autonomous-enabled mission command reference architectures?

5. Autonomy-Enabled Mission Command - Areas of Interest

In addition to industry input to the reference architecture, respondents are encouraged to provide information on system solutions for any or all of the following Autonomy-Enabled Mission Command areas of interest. Please clearly identify which areas your response addresses.

5.1. Autonomy-Enabled Mission Command Software

Figure 1 features a high-level description of a autonomous-enabled mission command reference architecture framework.

- Do you have a software product which fits well in this high-level description? If not, how would you define the boundary?

- Do you find clarity in our terminology associated with the categories featured in the figure: Mission Autonomy, Autonomy-Enabled Mission Command, Autonomous Planning and Orchestration, Autonomous Collaboration and Teaming, Autonomous Mobility, and Autonomous Effects?
- Would your system benefit from installing software directly on the platforms themselves rather than just the controller, e.g. can you perform distributed decision-making?
- Describe briefly the multi-agent behaviors you have implemented in your software.
- Is your solution resilient to network disruption? If so, to what degree?
- How scalable is your autonomous-enabled mission command reference architecture implementation?

5.2. Entity to Entity Autonomy and Intelligence

Individual entities within a multi-entity formation may inform the behaviors of associated entities to achieve improved overall performance of the formation. Topic areas of interest include:

- Shared perception systems (sensor fusion, object detection, threat assessment)
- Collective autonomous planning and decision-making algorithms
- Group adaptive learning capabilities and performance optimization
- Shared real-time path planning and obstacle avoidance in dynamic environments
- Shared Friend-Foe identification and targeting systems
- Collective energy management and mission endurance optimization

5.3. Multi-Entity Coordination

Multiple, heterogeneous entities working in conjunction with each other will introduce a variety of complexities may be addressed and opportunities that may be realized through autonomous-enabled mission command reference architecture functions. Topic areas of interest include:

- Task allocation algorithms for heterogeneous entity teams
- Formation control and deconfliction methods
- Distributed consensus and voting mechanisms
- Swarm/group intelligence and emergent behavior approaches
- Collective perception and shared situational awareness
- Dynamic team composition (entities joining/leaving missions)

5.4. Mission Management and Planning

Robust autonomous-enabled mission command reference architecture will allow efficiencies not previously achievable for unmanned systems; however, these advantages will only be realized with the implementation of high-level planning and interactions between individual entities.

Topic areas of interest include:

- Commander's intent interpretation and mission decomposition
- Automated mission planning from high-level objectives
- Real-time mission monitoring and execution tracking
- Adaptive replanning in response to contingencies
- Multi-objective optimization (time, resources, risk)

- Rules of Engagement (ROE) encoding and enforcement

5.5. Communications and Networking Relative to Autonomy-Enabled Mission Command

Autonomy-Enabled mission command orchestration of distributed entities must operate coherently in DDIL electro-magnetic spectrum. Topics of interest, as specifically relevant to commanding many unmanned systems, include:

- Resilient mobile ad-hoc network (MANET) protocols
- Delay-tolerant networking for intermittent connectivity
- Anti-jamming and low probability of detection (LPD) communications
- Adaptive bandwidth management and Quality of Service (QoS)
- Communication-denied operation modes
- Cross-domain information exchange (classification levels)

5.6. Human-Machine Interface and Control

The envisioned autonomous-enabled mission command reference architecture retains a human-in-the-loop or human-on-the-loop framework, wherein operators may interact with autonomous entities when and where required. Well-structured mechanisms for human interface and control are a necessity for robust autonomous-enabled mission command reference architecture. Topic areas of interest include:

- Multi-level command and control interfaces (strategic, operational, tactical)
- Visualization of complex multi-entity operations
- Alert prioritization and decision support systems
- Variable autonomy frameworks (adjustable human control)
- Explainable AI for autonomous decisions
- Workload management and cognitive load reduction

5.7. Safety, Security, and Assurance

Formation composed in whole or in part of autonomous entities will be held to many of the same requirements, and vulnerable to many of the same threats as conventional formations.

Forethought must be given to addressing these concerns. Topic areas of interest, as specifically relevant to commanding many unmanned systems, include:

- Assured control safety and security architectures (note related RFI in section 6.6)
- Cybersecurity architectures (defense-in-depth, zero-trust)
- Anti-spoofing and sensor validation techniques
- Formal verification methods for safety-critical functions
- Fault detection, isolation, and recovery (FDIR)
- Adversarial AI defense mechanisms
- Supply chain security for hardware and software components

5.8. Testing, Verification, and Validation

Seeking information on the technical maturity, testing and verification of solutions. Topics of interest include:

- Technology Readiness Level (TRL) for each component
- Current state of development and testing
- Simulation environments for multi-entity systems
- Hardware-in-the-loop (HITL) and software-in-the-loop (SITL) testing
- Digital twin approaches for system validation of multi-entity systems
- Metrics and evaluation frameworks for multi-entity autonomous systems
- Operational test and evaluation methodologies
- Certification approaches for multi-entity autonomous systems

5.9. Past Experience

Seeking description of past performance and experiences with the following topics of interest:

- Have you worked with the Army's Uncrewed Vehicle Control (UVC)?
- Have you controlled platforms which conform to RAS-A, MAVLink, RAS-G/IoP, STANAG 4586 standards, or other interface standards deployed on US military systems?
- If applicable, what lessons have you learned from integration with a military program of record vehicle or prior military demonstrations or operational deployments?

6. RFI Response Instructions

Responses should be submitted electronically to the point of contact listed in Section 8. All submissions must be received by April 9, 2026, at 5:00 PM Eastern Time.

Page Limit and Format

- Technical response: Maximum 15 pages (excluding cover page and appendices)
- Font: Arial or Times New Roman, 11-point minimum
- Format: PDF, Microsoft Word, or equivalent
- Margins: 1 inch on all sides

Cover Page: Submittals must include a Company or Entity cover page, limited to one (1) page, which will not count against the 15-page limit, and should include Company name and address, and CAGE Code (if available); the Technical and Business Points of Contact (POC) to include: Telephone Numbers and Email Addresses; Company headquarters location; size of company, both revenue and # of employees; if the Company is designated as a non-traditional or traditional contractor; if the Company's proposed solution/approach has been proposed to or previously funded by any federal agency, to include SBIR/STTR, please identify each specific agency and all applicable contract/agreement numbers; and a brief abstract (1,000 character limit which should be included in the appendix). Also state whether you are a member of NAMC or any other defense related consortium. The technical submittals should include the contribution to the requirement and provide a high-level project overview describing how this contribution addresses the stated requirements.

Classification: This RFI and all responses shall be UNCLASSIFIED. Do not include any classified, proprietary, or export-controlled information unless specifically requested and cleared through appropriate channels.

7. Important Notices and Disclaimers

7.1. Nature of this RFI

This RFI is issued for information and planning purposes only and does not constitute a Request for Proposal (RFP) or a promise to issue an RFP. This RFI does not commit the Government to contract for any supplies or services. Responses to this RFI are not offers and cannot be accepted by the Government to form a binding contract.

7.2. No Funding for Responses

Respondents are solely responsible for all expenses associated with responding to this RFI. The Government will not reimburse any costs incurred in preparing or submitting responses.

7.3. Proprietary Information

Respondents should clearly mark any proprietary or business-sensitive information. The Government acknowledges its obligations under the Federal Acquisition Regulation (18 U.S.C. §1905) to protect confidential information provided to the USG. Pursuant to this statute, the Government is willing to accept any trade secrets or proprietary restrictions placed upon data presented in response to this RFI and to protect such information from unauthorized disclosure. While the Government will make reasonable efforts to protect such information, responses may be subject to the Freedom of Information Act (FOIA). Respondents should not include information they wish to protect from public disclosure unless absolutely necessary.

7.4. Government Rights/Disclaimer

The USG is not responsible for the public disclosure of unmarked data received in response to this RFI. Accordingly, the potential offeror shall ensure that technical data provided in response to this RFI shall be appropriately marked (i.e., PROPRIETARY) to ensure proper Government handling of the data. Additionally, marked data that is already in the public domain or in the possession of the Government or third parties, or is afterward placed into the public domain by the owner or another party through no fault of the Government will not be protected once in the public domain. Data already in the possession of the Government will be protected in accordance with the USG's rights in the data. Should the Government need to reproduce the protected data for distribution purposes between Government offices, all such data shall be reproduced with restrictive legends in place. The Government may employ AI as a tool to assist in the analysis and review of RFI submissions. AI tools may be used to support tasks such as summarizing content, and highlighting areas of potential strength, weakness, or risk. The Government retains sole responsibility for all inherently governmental functions and will not delegate decision making authority to any AI tool. The AI tools utilized by the Government will be employed in a manner consistent with applicable regulations that govern the protection of proprietary and source selection sensitive information. The AI tool will be an output-only system. Any generated report data is not saved. No RFI data will be transmitted to or processed by external, non-

Government systems without appropriate safeguards and authorizations in place. USG support contractors may assist in reviewing RFI submissions. Information submitted in response to this RFI may be used by the Government for any purpose, including but not limited to: developing requirements, preparing solicitations, evaluating proposals, or sharing with other Government agencies. The Government may discuss information in responses with other potential contractors as part of market research activities.

7.5. Questions

If a vendor has questions, they shall be included as an appendix to the vendor's RFI response.

7.6. Related RFIs

CPE MA has released the related "Robotic Ground Combat Systems RFI" with NAMC via <https://www.namconsortium.org/opportunities/rfi-00-27-cpe-mission-autonomy> and "Landscape of Current Industry Robotic Solutions," with SAM.GOV via <https://sam.gov/workspace/contract/opp/f37847b6c34e42bdb4c2f9f835649594/view>. An RFI entitled "Safety Assured Control and Security Architecture for Mission Autonomy" is anticipated in the near future.

8. RFI Submission Instructions

All inquiries and responses should be directed to:

General questions : questions@namconsortium.org

RFI Response Submissions to: submissions@namconsortium.org

Subject Line: "RFI Response - MEAMC-2026-001 - [Company Name]"

RFI Response Deadline - April 9, 2026, at 5:00 PM ET

Maximum file size: 20 MB (Reach out to NAMC via questions@namconsortium.org if a larger submission is required)

A panel of Army and contractor SMEs will review submittals. It is expected that discussions may occur between Army SMEs and the contractor/performer for the purpose of clarifying technical submissions. All RFI submissions are treated as company proprietary information, and the content will be disclosed to U.S. Government (USG) employees, military, or designated support contractors only for the purpose of reviewing technical feasibility and market research. RFI submittals should outline both capacities and technical solutions/concepts and prototypes that address and support the stated objectives and requirements outline herein. The 15-page technical submittal, in contractor format, may focus on describing details of a proposed solution or prototype, how it could address the technical requirements, and potential impacts and benefits to the Army's Autonomous Mission.

9. Key Dates

RFI Release Date - March 11, 2026

RFI Response Deadline - April 9, 2026, at 5:00 PM ET

Note: All times are Eastern Time (ET). Late submissions will not be accepted unless explicitly authorized by the Government.